INCH-POUND

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DETAIL SPECIFICATION

CARTRIDGE, IMPULSE, MARK 19 MOD 0

This specification is approved for use by the Naval Surface Warfare Center Indian Head Division (NSWC IHD) and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the manufacture and acceptance of the Mark 19 Mod 0 impulse cartridge (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-D-21625 - Design and Evaluation of Cartridges for Cartridge Actuated Devices

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1168 - Ammunition Lot Numbering and Ammunition Data Card

DOD-STD-2101 - Classification of Characteristics

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL AIR SYSTEMS COMMAND (NAVAIR) DRAWINGS

1825934 - Igniter Assembly

2164465 - Cartridge, Impulse Mark 19 Mod 0 Assembly

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 1377

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2302165 - Loaded Spacer

DL419700 - Cartridge, Impulse, Mark 19 Mod 0

(Copies of these documents are available from Commander, Indian Head Division, Naval Surface Warfare Center, CAD/PAD Data Management Branch (Code E26), 4393 Benson Road, BLDG 1557, Indian Head, MD 20640-5092. Refer questions, inquires, or problems to nswc_iheodtd_dmo@navy.mil.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

OP 5 - Ammunition and Explosives Safety Ashore

(Copies of this document are available online via Model Based Product Support (MBPS) at https://mbps.navsea.cloud.navy.mil/Windchill. To gain access to MBPS, obtain an account with National Help Desk Service Management (NHDSM) at https://nhdsm.navair.navy.mil (a valid CAC is required to access this website) and submit a SAAR-N Request. Refer questions, inquiries, or problems to (888) 292-5919. This document is available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM E1742/E1742M - Standard Practice for Radiographic Examination

(Copies of this document are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Conformance to documents. The Mark 19 Mod 0 impulse cartridge covered by this specification shall be manufactured in accordance with Drawing DL419700 and all documents listed thereon.

3.3 Materials. All materials used in the manufacture of the cartridges shall conform strictly to the specifications referred to on the respective drawings.

3.4 Primary components. For the purposes of this specification, the propellant is considered a primary component.

3.5 Insulation resistance. Prior to the assembly of the bridgewires in the inner cup and electrode assembly, the insulation resistance shall be not less than 50 megohms at a relative humidity of less than 80 percent (see 4.5.1).

3.6 Bridgewire. The bridgewire shall be soldered to the inner cup and electrode. The solder and soldering process shall be as specified on the applicable drawing.

3.7 Bridgewire assembly, continuity. Before filling the cavity around the bridgewire with the igniter compound, the bridgewire assembly shall be thoroughly cleaned to remove solder flux and contamination. After cleaning, the bridgewire shall be flashed by application of an electric current to assure continuity of both legs of the bridgewire. The current shall be regulated to produce a visible red glow of the bridges. An electromagnetic force (EMF) of approximately 1.5 volts alternating current (AC) or direct current (DC) may be used with an appropriate series resistor to adapt the source characteristics to the primer resistance. Flashing is a necessary manufacturing operation and shall be conducted on each bridgewire assembly. Assemblies that fail to flow in both legs of the bridge shall be screened from acceptable material in the process. The discarded assemblies may be reworked by replacement of the bridge.

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3.8 Igniter composition and loading. The igniter composition shall be compounded and loaded in accordance with Drawing 1825934.

3.8.1 Loaded spacer. The loaded spacer shall be loaded with primer composition FA-878 in accordance with Drawing 2302165.

3.9 Propellant charge.

3.9.1 Quantity. All propellant charges shall be as specified on Drawing 2164465. Charges shall be determined by weight rather than by volume measurement.

3.9.2 Smokeless powder, handling, and loading. The smokeless powder shall be subjected to no unnecessary handling or exposure to the atmosphere. The handling and loading of smokeless powder exposed to the atmosphere shall be performed only under conditions that do not permit the relative humidity to exceed 75 percent.

3.9.3 Smokeless powder, condition. The smokeless powder used shall exhibit no evidence of decomposition (see 4.5.2).

3.10 Cartridge requirements.

3.10.1 Bridgewire resistance. The resistance of the bridge circuit in each completely assembled cartridge, when measured electrically (see 4.5.4), shall be not less than 0.08 ohm nor more than 0.14 ohm.

3.10.2 Leakage. Cartridge leakage rate shall be not greater than 10^{-5} cubic centimeters of air per second (see 4.5.5).

3.10.3 Radiographic examination. Cartridges shall be free of imperfections when in components or assembly as verified by radiographic examination of all first article and production test samples (see 4.5.6).

3.11 Ballistic requirements. Cartridges shall meet the ballistic requirements specified in 3.11.1 through 3.11.3 when subjected to the test prescribed in 4.5.7.

3.11.1 Chamber pressure. The upper and lower limits of maximum pressure in pounds per square inch shall be 2,400 and 1,550, respectively.

3.11.2 Misfire. There shall be no misfires (see 4.5.7.2).

3.11.3 Hangfire. There shall be no hangfires (see 4.5.7.3 and 6.5).

3.12 Change in propellant lot. Whenever a change occurs in the propellant lot for the main charge, test firings shall be conducted in accordance with 4.7. No cartridge shall produce results outside the limits specified in 3.11.1, and no cartridge shall fail to meet the requirements of 3.11.2 or 3.11.3.

3.13 Defects. Cartridges shall be free of the following defects: burrs, dents, incomplete or damaged threads, missing or damaged O-ring, perforated closure disc, or other defects that may prevent entry of the cartridge into the firing chamber or affect the resistance of the cartridge to moisture. All cartridge marking and identification shall be verified using MIL-STD-1168 (see 4.5.3).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Conformance inspection (see 4.4).

4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the applicable test methods of 4.5.

4.3 First article inspection. First article inspection shall be performed in accordance with 3.1. A first article sample acceptable for environmental and function testing shall be considered as a sample that has met the requirements of 3.2 through 3.10.3 and 3.13 of this specification. First article inspection shall be performed on sample units that have been produced using the same materials, equipment, processes, and procedures as will be used in regular production. All parts and materials shall be the same as used for regular production and shall be from the intended sources of supply.

4.3.1 Sample size. The first article sample shall consist of 94 cartridges.

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4.3.2 Inspection routine. Eighty-four cartridges shall be expended in the tests listed below and 10 shall be retained for investigative purposes. Testing shall consist of the following tests of MIL-D-21625 (the number in parentheses denotes the number of cartridges allocated for each test): 40-foot drop test (6); 15 “g” shock (12); Temperature and Humidity Cycling (THC) (18); vibration (12); -65 °F temperature (12); +160 °F temperature (12); +70 °F temperature (12). The cartridges shall be fired in the test set specified in 4.5.7. No cartridge firing shall produce results outside the limits specified in 3.11.1 through 3.11.3.

4.3.3 Failure. Failure of any cartridge shall be cause for rejection of the first article sample.

4.4 Conformance inspection. Conformance inspection shall consist of the tests specified in table I.

TABLE I. Conformance tests and examinations.

Classification item	Nature of test	Requirement paragraph	Test paragraph	Classification (DOD-STD-2101)
Insulation	Resistance	3.5	4.5.1	(M101)
Propellant charge	Quantity	3.9.1	-	(M102)
Smokeless powder	Condition	3.9.3	4.5.2	(M103)
Cartridge	Visual examination	3.13	4.5.3	(M104)
Cartridge	Bridgewire resistance	3.10.1	4.5.4	(M105)
Cartridge	Leakage	3.10.2	4.5.5	(M106)
Cartridge	X-ray	3.10.3	4.5.6	(M107)
Cartridge	Ballistics	3.11	4.5.7	(M108)

4.4.1 Lot size. Unless otherwise specified (see 6.2), the production lot size shall be not less than 1,301 and not greater than 8,000 maximum. Only primary components from a single lot shall be used in a production lot of cartridges; however, one primary component production lot may be used in more than one cartridge production lot.

4.4.2 Sampling. Unless otherwise specified (see 6.2), sampling shall be in accordance with table II.

TABLE II. Conformance sample size.

Lot size	Sample size	Retained sample
66-110	15	10
111-180	15	10
181-300	20	10
301-500	25	10
501-800	30	10
801-1,300	35	10
1,301-3,200	40	10
3,201-8,000	50	10
8,001-22,000	75	10

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4.4.3 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as critical, major, or minor in accordance with DOD-STD-2101. Tests and examinations that verify critical characteristics are identified by the symbol (C) and major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of the test or examination. Tests and examinations that are not annotated with a classification code are classified minor.

4.5 Tests.

4.5.1 Insulation resistance. Prior to assembly of the bridgewire in the inner cup and electrode assembly, the insulation resistance shall be checked with a 500-volt megohmmeter. Units with resistance of less than 50 megohms shall be cause for rejection.

4.5.2 Smokeless powder, condition. Smokeless powder shall be inspected in accordance with the applicable paragraphs of OP 5. Evidence of decomposition shall be cause for rejection.

4.5.3 Visual examination. All cartridges shall be visually examined. Cartridges having any of the defects of 3.13 shall be cause for rejection.

4.5.4 Bridgewire resistance. Conformance with the bridge circuit resistance of 3.10.1 shall be ascertained on all completely assembled cartridges by means of a test circuit that limits the bridge circuit current to 50 milliamperes, maximum. A safety chamber with an interlock switch shall be used to protect the operator during the application of current. An instrument that is known to be accurate to within 0.005 ohm in the range of resistance specified shall be used to determine whether or not the requirement is being met. Units that fail to meet the requirements of 3.10.1 shall be cause for rejection. The results of bridge circuit resistance measurements shall not be affected by resistance of the test circuit and connector.

4.5.5 Leakage. Each cartridge in the first article or production sample shall be leak tested in a dry gas leak tester. Cartridges that exhibit a leak rate greater than 10^{-5} cubic centimeters per second of air at a pressure differential of 1±0.1 atmosphere shall be considered defective. Cartridges of the first article sample that exhibit a leakage defect shall be cause for rejection. If cartridges of the production lot sample exhibit a leakage defect (see 6.4), then a 100 percent leak test of the lot being tested shall be required. Cartridges that exhibit a leak rate in excess of that specified above shall be cause for rejection (see 6.6).

4.5.6 Radiographic examination. Cartridges for ballistic testing shall be radiographically examined in accordance with ASTM E1742/E1742M. Any observable imperfections in components or assembly shall be cause for rejection (see 3.10.3). Failure to meet sample requirements shall result in 100 percent radiographic screening of the production lot.

4.5.7 Ballistic tests. The number of cartridges selected from each production lot for the ballistic test shall be in accordance with table II. The cartridges shall be conditioned for 24 hours at 70±5 °F and fired in a closed bomb test fixture having a volume of 2.7 cubic inches (including the cartridge volume). The chamber shall be thoroughly cleaned before each firing and the cartridge temperature shall be 70 °F. A record of pressure versus time (i.e., time from application of the firing current) shall be taken for each firing. Firing current shall be 10 amperes. Acceptance criteria (regardless of lot size) shall be: A failure of any cartridge to meet this requirement shall be cause for rejection of the lot represented.

4.5.7.1 Criteria of acceptability. The upper and lower limits of maximum pressure shall meet the requirements of 3.11.1. A failure of any cartridge to meet this requirement shall be cause for rejection of the lot represented.

4.5.7.2 Misfire. Unless otherwise specified (see 6.2), the failure of any cartridge to fire shall be cause for rejection of the lot represented, unless such failure is plainly attributable to faulty test equipment or procedure.

4.5.7.3 Hangfire. One or more hangfires shall be cause for rejection of the lot represented.

4.6 Retest. There shall be no retests, except as specified in 4.6.1.

4.6.1 Test failure. If test failure is attributable to an assignable cause, excluding the test cartridge, original test results should be discarded and that portion of the test reconducted.

4.7 Change in propellant lot. Whenever a change in propellant lot for the main charge occurs, the charge weight shall be verified by firing 36 cartridges that have been temperature conditioned as follows: 12 at -65±5 °F, 12 at 70±5 °F, and 12 at 160±5 °F. Results of these firings shall meet the requirements of 3.12.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Mark 19 Mod 0 impulse cartridge is intended primarily as an aircraft stores separation cartridge (replacing the Mark 25 primer). As the impulse cartridge is designed only for military aircraft, there is no commercial application for the Mark 19 Mod 0 impulse cartridge. Current stock of Mark 19 Mod 0 may be used until depleted; future requisitions for Mark 19 Mod 0 should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When first article inspection is required (see 3.1, 4.3, and 6.3), and if so, specify the test activity.

c. Inspection conditions, if other than as specified (see 4.2).

d. That revision "G" of MIL-D-21625 be used (see 4.3.2).

e. Lot size, if other than as specified (see 4.4.1).

f. Sampling, if other than as specified (see 4.4.2).

g. Packaging requirements (see 5.1).

h. That the safety precaution requirements of DoD Manual 4145.26, "DoD Contractor's Safety Manual For Ammunition and Explosives" are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of OP 5, "Ammunition and Explosives Safety Ashore", are applicable.

6.3 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of 94 cartridges. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product that has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Leakage. An AQL of 2.5 percent defective from ANSI/ASQ Z1.4 has been used successfully in the past for purposes of the number of cartridges exhibiting leak defects during production lot sample leak test and should be seriously considered when writing the contract or purchase order.

6.5 Hangfire definition. A hangfire is defined as a firing in which the elapsed time between actuation of the firing mechanism and case opening or start of pressure rise exceeds 0.05 second.

6.6 Advisory leak test. Indian Head leak tests cartridges should be in accordance with AD-783 943, "Cartridge Leak Test Procedures at the Naval Ordnance Station, Indian Head, Maryland."

6.7 Subject term (key word) listing.

Separation
Stores

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Preparing activity:
Navy – OS
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